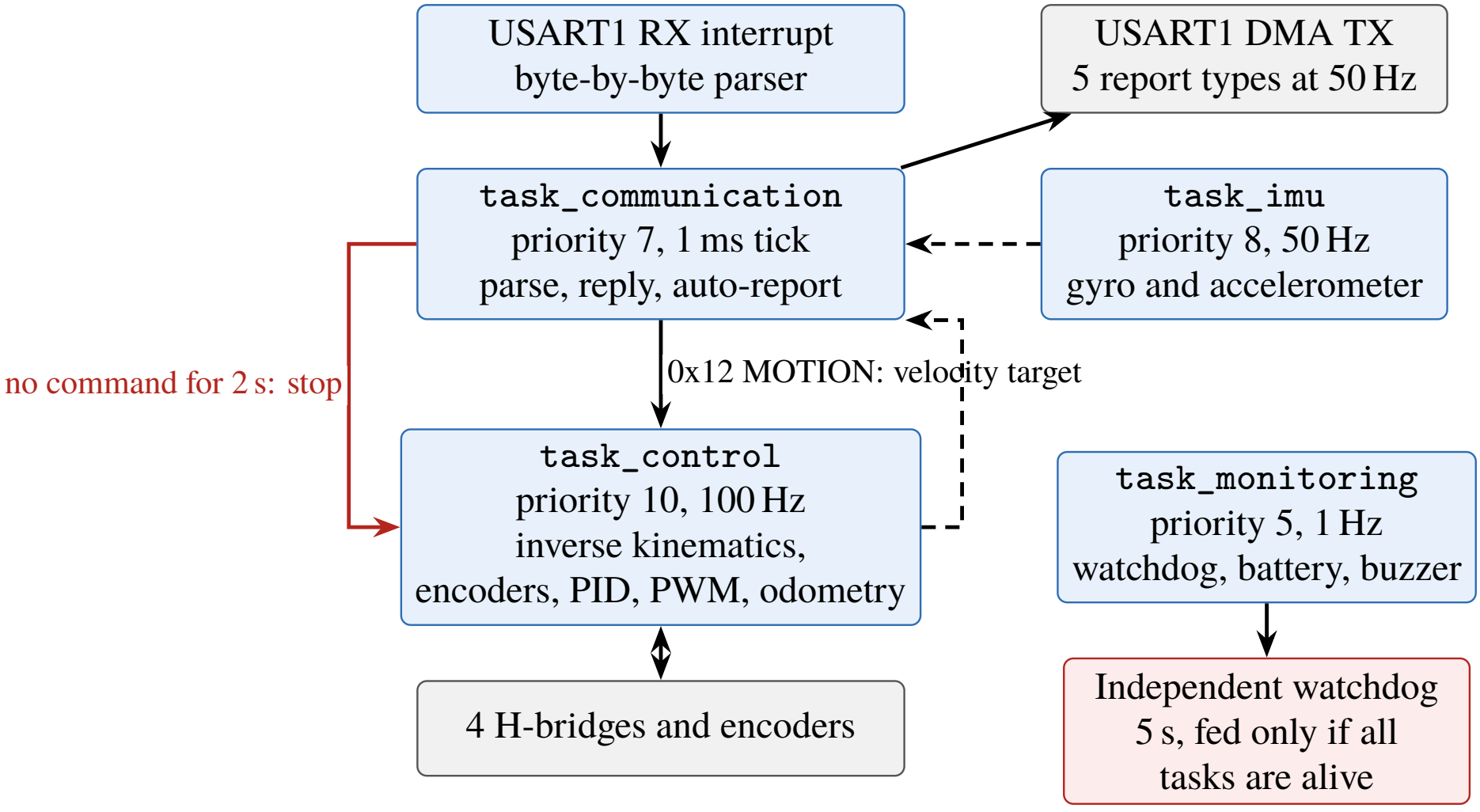
USART1 RX interrupt
byte-by-byte parser
USART1 DMA TX
5 report types at 50 Hz
task_communication
priority 7, 1 ms tick
parse, reply, auto-report
task_imu
priority 8, 50 Hz
gyro and accelerometer
no command for 2 s: stop
0x12 MOTION: velocity target
task_control
priority 10, 100 Hz
inverse kinematics,
encoders, PID, PWM, odometry
task_monitoring
priority 5, 1 Hz
watchdog, battery, buzzer
4 H-bridges and encoders
Independent watchdog
5 s, fed only if all
tasks are alive